

CloudEagle - DevOps Assignment

Part 1 – Deployment & CI/CD Design

Scenario

We have a **Spring Boot backend service** (`sync-service`) that:

- Connects to MongoDB
- Is deployed to GCP VMs
- Has environments: `qa`, `staging`, `prod`
- Is built via Jenkins

Your Task

Design a **CI/CD pipeline** for this service.

You should cover:

1. **Branching strategy**
 - How do branches map to environments?
 - How do you avoid accidental prod deployments?
2. **Jenkins pipeline**
 - High-level pipeline stages
 - What happens on PR vs merge?
 - Rollback strategy if deployment fails
3. **Configuration management**
 - How do you manage env-specific configs?
 - Secrets handling (MongoDB creds, API keys)
4. **Deployment strategy**
 - Blue/Green vs Rolling vs Recreate (justify)
 - Zero/minimal downtime approach



Deliverable - Share a Github project with the following

- A short design document (Markdown or PDF)
- Jenkinsfile

Part 2 – Infrastructure Design

Scenario

We want to run this service on **GCP** with:

- Auto-scaling
- Secure access
- Reasonable cost (startup constraints)

Your Task

Propose an infrastructure setup.

You should touch on:

- Compute choice (GKE / Compute Engine / Cloud Run — justify)
- MongoDB hosting approach
- Networking basics (VPC, ingress)
- Secrets & IAM
- Logging & monitoring stack

 **Deliverable - Share a Github project with the following**

- Architecture diagram
- Written explanation of key choices